Syllabus Sync

University Integration & Security Requirements

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For:	Macquarie University IT Services

1. Purpose

Syllabus Sync is a student-built mobile-first web application that helps Macquarie University students manage their academic schedules, track deadlines, discover campus events, and navigate the campus map all in one place. The app is designed around the student lifecycle at MQ and is being prepared for a public launch at **Open Day in August 2026**.

This document outlines what the development team needs from the University to move from a working prototype to a production-quality, university-endorsed application. The goal is to identify the systems, access, and approvals required and to start those conversations early enough to meet the August timeline.

2. Required University Systems & APIs

The following integrations would enable Syllabus Sync to display live, accurate data rather than manually entered information.

2.1 Timetable / eStudent Data

What we need: Read-only access to a student's enrolled units and their class schedule (unit code, name, day/time, room, building).

Why: Currently, students must manually add their units and class times. If we could pull this from eStudent or the timetable system, the app would be pre-populated on first login — significantly reducing friction and improving accuracy.

Ideal format: REST API or structured data export (JSON/CSV) per authenticated student.

2.2 iLearn / Assessment Data

What we need: Read-only access to upcoming assessment dates, types, and weightings for enrolled units.

Why: Deadline tracking is a core feature. Auto-importing assessment dates from iLearn would mean students always have up-to-date deadlines without manual entry and would reduce the risk of missed submissions.

Ideal format: LTI integration or an API endpoint that returns assessment metadata per unit.

2.3 Campus Events Feed

What we need: Access to the university's official events calendar (student events, career fairs, orientation activities, faculty events).

Why: Syllabus Sync has an Events tab that students use to discover campus activities. A live feed from the university's events system would keep this up to date and authoritative.

Ideal format: iCal feed, RSS, or REST API. Even a public URL we can scrape periodically would be a starting point.

2.4 Campus Map & Building Data

What we need: Official building metadata (names, codes, coordinates, accessibility info) and any wayfinding data.

Why: The app includes a full campus map with building search and navigation. We currently maintain our own building database (130+ entries sourced from OpenStreetMap), but official data would improve accuracy.

Ideal format: GeoJSON, CSV, or access to the campus GIS system.

3. Authentication & Single Sign-On

Current State: The app currently uses Supabase Auth with email/password authentication, WebAuthn (passkeys), and optional MFA. Email verification is enforced via Resend.

Requested Integration: We would like to integrate with Macquarie University's SSO (Single Sign-On) system so that students can log in with their MQ credentials.

Requirements:
• Support for SAML 2.0 or OpenID Connect (OIDC) — either protocol works with our stack
• A registered client/application in the university's identity provider
• Scope: authenticated MQ students (and optionally staff)
• Claims/attributes needed: email, full name, student ID, faculty (if available)

Benefits to MQ:
• No separate credentials for students to manage
• University retains control of authentication and account lifecycle
• Session policies (timeout, MFA) can be enforced centrally

4. Security Requirements

4.1 Application Security (Implemented)

• **HTTPS everywhere** — all traffic encrypted via TLS 1.3 (enforced by Vercel)
• **Row-Level Security (RLS)** — every database table is protected; users can only access their own data
• **CSRF/XSS protection** — enforced by Next.js defaults and Content Security Policy headers
• **Rate limiting** — server-side rate limiting on all API endpoints and authentication flows
• **Input validation** — Zod schema validation on all user inputs (client and server)
• **Audit logging** — all sensitive operations logged with IP, user agent, and timestamps; 90-day retention

4.2 Requested from MQ

• **Security review/penetration test:** We welcome a university-commissioned penetration test before any official endorsement or SSO integration.
• **Privacy Impact Assessment (PIA):** Guidance on conducting a PIA under the Privacy and Personal Information Protection Act 1998 (NSW).
• **Data classification guidance:** Confirmation of the classification level for student schedule and assessment data.

4.3 Data Handling

• **Storage:** All user data stored in Supabase (AWS ap-southeast-2, Sydney region)
• **Encryption:** Data encrypted at rest (AES-256) and in transit (TLS 1.3)
• **Data minimisation:** We only store what the student explicitly provides
• **Right to deletion:** Users can delete their account and all associated data via the app
• **No data sharing:** Student data is not shared with third parties or advertisers

5. Infrastructure & Hosting

5.1 Current Stack

Component	Provider	Region
Frontend + API	Vercel	Sydney (syd1)
Database	Supabase	AWS ap-southeast-2
Email	Resend	US (sender domain pending)
Maps	OpenStreetMap + Leaflet	Self-hosted tiles

5.2 What We Need from MQ

1. **Custom domain approval:** Serve the app from a *.mq.edu.au subdomain (e.g., syllabus-sync.mq.edu.au). This requires a DNS CNAME record.

2. **Sender domain for email:** Send emails from an @mq.edu.au address. Requires MX/SPF/DKIM records.

3. **Push notification credentials:** Web push notifications support (or integration with existing MQ platform).

4. **Production environment variables:** If SSO is approved, we need the IdP metadata URL, client ID, and client secret.

6. Risks & Mitigations

Risk	Impact	Mitigation
API access delayed	Demo relies on manual data	Maintain high-quality seed data; compelling demo follow
SSO integration timeline	Students can't use MQ credentials	Keep email/password + passkey auth as fallback
Data accuracy	Timetable data may be stale	Display 'last synced' timestamps; allow manual overrides
Availability	App downtime during Open Day	Vercel 99.99% SLA; load-test before event
Privacy concerns	University hesitant to endorse	Conduct PIA early; vendor risk assessment

7. Proposed Timeline

Phase	Timeframe	Milestones
Now	March 2026	Demo account ready; requirements shared; initial meeting requested
Discovery	April 2026	Meeting with IT Services; identify API contacts; begin SSO process
Development	May–June 2026	Build API integrations; SSO integration; security hardening
Access & Testing	Early July 2026	API credentials received; end-to-end testing; pen test
Staging	Mid-July 2026	Staging on *.mq.edu.au subdomain; UAT with select students
Launch Prep	Late July 2026	Load testing; final security review; marketing materials
Open Day	August 2026	Public demo; QR codes for student sign-up; live onboarding

Critical path: API access and SSO credentials by early July are essential for a confident Open Day launch. Without these, we can still demo the app with seed data, but the "wow factor" of live data will be missing.

8. Next Steps

1. Schedule a meeting with Richard and the IT Services team to discuss feasibility and process.

2. Submit a formal application for API access (timetable, events, assessment data).

3. Begin the SSO client registration process.

4. Share this document with relevant stakeholders for feedback.

This document is a draft prepared by the Syllabus Sync development team (Pouya Alavi & Raouf Abedini). It is intended to start a conversation, not to prescribe solutions. We are open to alternative approaches and welcome guidance from the university's IT and security teams.